Regional Dimensions of Migration Sentiments in France: An NLP Analysis of Media Coverage Following the Charlie Hebdo Attack

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Research Question

This analysis explores how **French media perceptions of immigration** evolved over a **five-year time window**, particularly in response to the **2015 Charlie Hebdo attack**. Using sentiment analysis in a **Natural Language Processing (NLP) model**, we examine variations in migration narratives across regions and time, analyzing **31,255 newspaper articles** scraped from Europresse [1] to uncover how national shocks and local contexts shape public attitudes toward migration. The study employs both **long-term and short-term perspectives**, using two distinct time windows for analysis.

Why We Chose This Question?

In the current context of rising polarization about immigration in Europe, studying media representations offers valuable insights into how narratives influence integration, prejudice, and policy formulation, making it a vital area of inquiry for both academic and societal progress.

Gehring et al.(2022) [2] explore how media narratives about immigrants shape public sentiment and perceptions. It highlights the importance of understanding narratives as they influence societal, economic, and political decisions, such as voter behavior and public attitudes toward immigration policies. Their work provides a structured framework for classifying immigrant-related narratives in newspaper articles into predefined themes, such as economy, cultural integration, or religion, while also assigning sentiment to these narratives. This work inspires our research on the sentiment perception of migrants in the French context, as it underscores the critical role media narratives play in influencing societal attitudes. Using **83 French regional newspapers** available in Europresse, we constructed a database of **31,255 articles** to analyze how immigrants are represented and the sentiment attached to these representations.

Through cross-regional and two-time-period analysis, we investigate **how media narratives vary over time and between regions, and how national shocks, such as terrorist attacks, differently influence these spatial media representations in the long and short-term.**

Data and Descriptive Statistics

We initiated our data extraction using the Europresse website, initially dividing France into **seven predefined regions**. After France's administrative reform in January 2015, we reorganized pre-2015 regions to match the new divisions (see Table A1 in the Appendix). For Île-de-France, only *Le Parisien* was available, which, as a national newspaper, may introduce bias compared to regions analyzed only by regional media. We used the keyword **"immigration*"** to focus our analysis.

Our study focuses on media coverage from June 21, 2012, to December 31, 2016, deliberately avoiding the period of strong French presidential campaign. To further investigate the specific impact of the Charlie Hebdo attack, we conducted a second analysis using a **restricted time window**, from December 6, 2014, to February 7, 2015. This period includes one month on either side of the event to isolate the shock of the attack while minimizing the influence of other influential events. However, the smaller time window resulted in greater variance, reducing precision compared to the broader analysis but it allows us to reduce the bias of unrelated events. An explanation of the database's construction is provided in the `Appendix`, as well as a guideline to replicate this construction in the `Read_Me_BDOS`.

Region	Before	After	Before Restricted Window	After Restricted Window
Center	1483	1754	56	63
Ile-de-France	1038	1133	55	49
North-East	2087	2647	65	102
North-West	1353	2046	53	45
North	2346	2641	106	103
South-East	2380	2838	83	116
South-West	2707	4802	137	224
Total	**13394**	**17861**	**555**	**702**

Table 1. Number of articles per period and per region

Table 1 provides an overview of the amount of articles we collected, per region and period. This allows us to contextualize our analysis by considering the number of articles in each region in the overall sentiment analysis. Specifically, we identified **13,394 articles** for the pre-period and **17,861 articles** for the post-period. For the restricted time window, the totals were **555 articles** for the pre-period and **702 articles** for the post-period. You will find the table of the number of newspapers per region and periods in the appendix (Table A2).

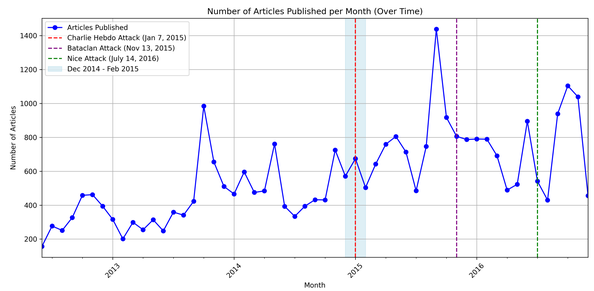

Figure 1. Articles published monthly from pre to post 2015 benchmarks in France

Figure 1 shows the monthly article count for the entire country over the broad time window, with the restricted window highlighted in blue. The dashed lines mark significant events: the Charlie Hebdo attack (red), the Bataclan attack (purple), and the Nice attack (green). We observed a slight increase in January 2015 following the Charlie Hebdo attack, but no significant spike occurs during the event month itself. A notable peak appears in September and October 2015, likely driven by the **2015 European refugee crisis**, which dominated media narratives during that period.

Model and approaches

In our study we aim to analyze and capture the main language sentiments expressed in the media across regions, both before and after our benchmark year. To achieve this, we deployed **two distinct NLP techniques** to conduct both abstract-level and headline-level analyses across a specified time period (restricted or not window) and spatial region.

First, we generated **WordClouds** to visualize the most frequently cited words in the entire database, combining abstracts and titles. Then, for each headline and each abstract, we used the **French VADER** (Valence-Aware Dictionary and Sentiment Reasoner) to compute **sentiment scores** categorized as positive, negative or neutral and a compound that combines these three scores into a single metric. In the appendix is the graphic representation of sentiment distribution (Figure A1). By combining these methods, our analysis captures both the dominant language patterns and the sentiment dynamics across different periods and regions.

Results and Figures

The WordClouds provide a visual representation of the most prominent words across our entire Europress database, highlighting the topics that were most prevalent in French regional newspapers.

Figure 2 and Figure 3 illustrate the prominents words across a larger period. Before the shock, immigration coverage was balancing political debate with human-centered narratives, with mentions of political figures and parties like "Manuel Valls" and the "Front National" alongside topics like family reunification and youth integration. For the post-January 2015 period (Figure 3), the narrative remained politically charged, with words like "L'Union Européenne," "Nicolas Sarkozy," and "Donald Trump," reflecting a **stronger right-leaning focus**. Additionally, terms like "Calais" and "réfugié" indicate growing attention to immigration challenges, also influenced by the 2015 European refugee crisis. This comparison highlights a shift in media narratives with deeper immigration challenges associated with prominent political figures.

Figure 4 and Figure 5 illustrate a clear shift in narratives in the period surrounding the Charlie Hebdo attack. In the month preceding the attack (Figure 4), media narratives on immigration were dispersed, with no dominant themes. Conversely, the month following the Charlie Hebdo attack (Figure 5) shows a significant shift. Words like "Paris", "attentat", or "Charlie Hebdo" become central, dominating the narrative. It reflects how media coverage of immigration focused on the attack, connecting it to **security concerns**. The national-level analysis of regional media narratives reveals that, in the long term, the Charlie Hebdo attack did not significantly change public discourse on immigration. However, a slight shift toward more challenging issues, likely driven by the 2015 European migration crisis, reflects the bias of a large sample. In the short term, as the last two figures illustrate, the attack reframed immigration-related discussions in the media narrative.

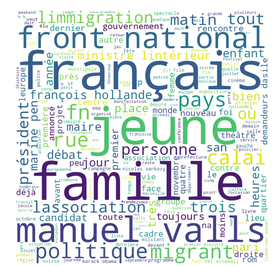

Figure 2. WordCloud Pre-2015

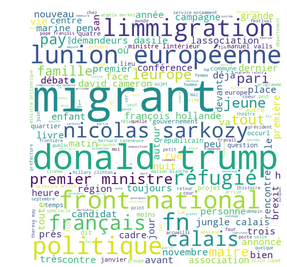

Figure 3. WordCloud Post-2015

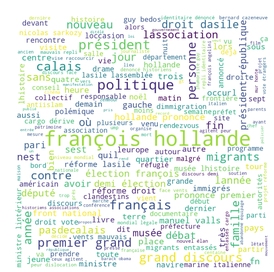

Figure 4. WordCloud Pre-Restricted

Figure 5. WordCloud Post-Restricted

Figure 6 and Figure 7 illustrate regional variations in sentiment across France before and after January 7, 2015. Figure 6 highlights a concentration of **negative sentiment in Île-de-France**, particularly in Paris, as shown by the deep red shading, as well as in **the South-West**. This pattern reflects the heightened intensity of negative media narratives surrounding migration in the capital, while other regions exhibited more moderate or even positive sentiment levels.

In contrast, Figure 7 illustrates the post-attack sentiment, revealing a more uniform distribution across the country. Notably, the intensity of negative sentiment in Île-de-France and the South-West diminishes, suggesting a shift toward greater national convergence. This change points to a potential collective societal response to the event, with public opinion and media discourse becoming less regionally polarized and more consistent nationwide.

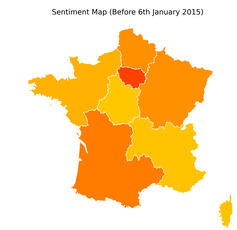

Figure 6. Pre-2015 France

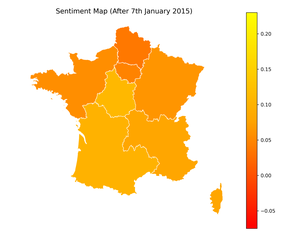

Figure 7. Post-2015 France

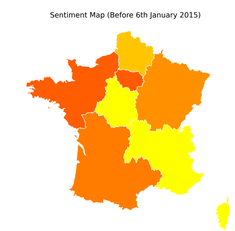

Figure 8. Pre-Restricted France

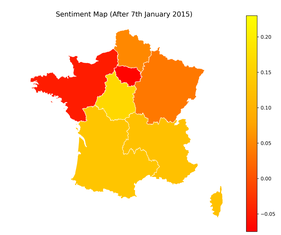

Figure 9. Post-Restricted France

Note: we rescaled the maps with the minimum to the maximum value (sentiment score) for all regions and periods

We analyzed shorter time windows to capture the heightened intensity of sentiment dynamics around the Charlie Hebdo attack. This approach revealed **sharper regional variations**, with notably more negative sentiment in the North-West and Île-de-France regions, reflecting the concentrated impact of the event. The short-term analysis also showed an **increased polarization**, with higher variance and extreme sentiment peaks, **highlighting a divergence in regional media narratives**. These findings demonstrate how national shocks like the attack shape media perceptions of immigration and amplify regional disparities in the short term.

Conclusion

We analyzed media framing of immigration in France by scraping 31,255 articles from 83 regional newspapers from Europresse, between 2012 and 2016, focusing on the impact of the Charlie Hebdo attack. Our study captured both short-term and long-term changes, revealing shifts in media narratives that linked immigration to security concerns and highlighted the role of current events in shaping public discourse.

Regional disparities emerged, with Île-de-France and the South-West showing pronounced differences in sentiment intensities. However, sentiment analysis is influenced by many factors, including regional factors, which can also explain variations in sentiment scores over this period, highlighting the need for a nuanced approach when interpreting media narratives.

Our research opens up new areas of investigation on how national crises reshape media representations of immigration, and on the implications of media framing for public policy and societal cohesion.

References

[1] Europresse.
Europresse, n.d.
Accessed on the November 5, 2024.

[2] Kai Gehring, Joop Age Harm Adema, and Panu Poutvaara.
Immigrant narratives.
2022.

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